



SCID-Compress900: A Multi-Scene Dataset of 4K and 1080P Screen Content Images for Image Compression Research

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Abstract

With the rapid growth of digital content and the increasing demand for high-resolution displays, the efficient compression of screen content images characterized by text, graphics, and UI elements has become an important research field. This paper introduces a new dataset named SCID-Compress900 specially designed for image compression research. The dataset consists of 900 high-quality screen content images, including 500 4K images and 400 1080P images. All these images are mainly composed of text/graphics, reflecting typical screen content scenarios such as office documents, software interfaces, and presentation slides. The dataset covers a diverse range of content, including various font sizes, graphic styles, and color modes, providing a comprehensive testbed for compression algorithms. To demonstrate the effectiveness of SCID-Compress900, we conduct benchmark tests using several deep learning-based image compression methods commonly employed by researchers. The experimental results show that SCID-Compress900 can well differentiate the performance of different compression algorithms. Compared with existing datasets, SCID-Compress900 offers higher resolution, larger scale, and more targeted content, making it an ideal resource for developing and evaluating advanced image compression algorithms for screen content. This dataset will not only promote the research and development of screen content compression technology but also contribute to the standardization and optimization of compression

algorithms in practical applications. The project is available at <https://openi.pcl.ac.cn/OpenDatasets/SCID-Compress900>.

CCS Concepts

• Theory of computation → Data compression.

Keywords

Screen Content Image, Image Compression, Benchmarking Test, Open-source Dataset, Deep Learning

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1 Introduction

In the era of digital information explosion, with the popularization of high-definition display devices such as 4K and 8K monitors, the number of screen content images (SCIs) generated in daily life has significantly increased. These contents cover office documents, video conference screenshots, software user interfaces, and other scenarios. Efficient compression of screen content images is crucial for reducing storage costs, accelerating transmission speeds, and enhancing the user experience of various applications such as cloud document sharing and real-time remote desktop services. Significant advancements have been made in related compression fields, including natural image compression [5, 11, 21], video compression [19, 20], and point cloud compression [13–15, 22]. However, screen content images have unique characteristics such as sharp text edges, repetitive patterns, and a high proportion of artificial graphics, which pose entirely different challenges compared to natural images. This makes the development of specialized compression algorithms critical.

The existing screen content image compression technologies mainly fall into two directions: traditional codec algorithms and deep learning-based methods. Traditional codec algorithm, such as HEVC Screen Content Coding (HEVC-SCC) [16], optimizes the structural features of screen content like text regions and graphics by introducing tools such as intra-block copy [17] and palette coding [10]. Meanwhile, deep learning-based methods [4, 6, 23] can analyze the features of screen content images to a greater extent and

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Figure 1: Overview of the SCID-Compress900. The figure shows representative image for each category of SCID-Compress900.

efficiently remove redundancy through effective network structure design, achieving higher compression ratios while demonstrating better visual quality. However, the performance evaluation of these algorithms highly depends on comprehensive datasets that can accurately reflect real-world screen content scenarios.

There is a significant structural contradiction in the current research status of screen content image datasets. At present, most existing screen content image datasets [7, 9, 18] are primarily designed for the field of image quality assessment, aiming to explore subjective or objective quality scores of compressed images. However, such datasets generally suffer from two major limitations: First, their resolutions are relatively low, mostly concentrated at 1080P or even lower, making it difficult to meet the demand for high-resolution data from emerging high-definition display devices (e.g., 4K, 8K) and compression standards. Second, they overly focus on providing distorted images processed by different methods, while the number of original high-quality source images is severely insufficient. This not only fails to support the requirements for the scale and diversity of original data in the development of compression algorithms but also cannot satisfy the benchmark test scenarios needed for algorithm performance comparison. In contrast, screen content image datasets specifically tailored for compression algorithm development are extremely scarce, which seriously hinders the development and fair comparison of advanced compression algorithms for high-definition screen content images.

To fill these gaps, we introduce SCID-Compress900, a new dataset specifically designed for screen content image compression research. SCID-Compress900 contains 900 high-fidelity screen content images. All images are carefully selected with text/graphics as the dominant content, covering a wide range of applications such as video conferences, programming interfaces, and online maps. The dataset is constructed using lossless acquisition to ensure the authenticity and high quality of the content, making it a reliable benchmark for evaluating screen content image compression algorithms. In summary, our contributions are as follows:

- We propose SCID-Compress900, a groundbreaking screen content image compression dataset. The dataset contains

900 high-fidelity images, focusing on text/graphics-rich content. Through a proprietary lossless acquisition protocol and strict quality control, we provide a dataset that significantly expands the scope of existing high-resolution algorithm development resources.

- We conduct a comprehensive analysis of SCID-Compress900, revealing its characteristics from multiple dimensions such as resolution, content type, and visual features. These analyses provide an empirical basis for understanding the unique patterns of screen content images and offer key references for developing compression algorithms adapted to their characteristics.
- We perform rigorous testing on various deep learning-based compression methods using SCID-Compress900, systematically evaluating key metrics such as PSNR and MS-SSIM, and establishing a reliable performance benchmark.

2 Dataset Construction

The data collection phase covers multiple types of typical screen content scenarios to ensure the diversity of the dataset. We collect original images from scenarios such as commercial presentation documents, integrated development environment (IDE) interfaces, web browsing interfaces, and professional drawing software interfaces through professional screen capture equipment and software. The collection sources cover commonly used software such as Microsoft Office, Matlab, and Pycharm, effectively avoiding content bias from a single platform or software. Importantly, all images are collected in a lossless manner to ensure the integrity and accuracy of the original content.

To meet the development needs of current mainstream display devices and compression technologies, the dataset adopts a hierarchical strategy in resolution planning: The 1080P resolution includes 400 images, subdivided into 3 categories:

- **Code** category (100 images): Focuses on programming code interfaces.
- **EdrawMax** category (200 images): Centers on professional drawing software operation interfaces.

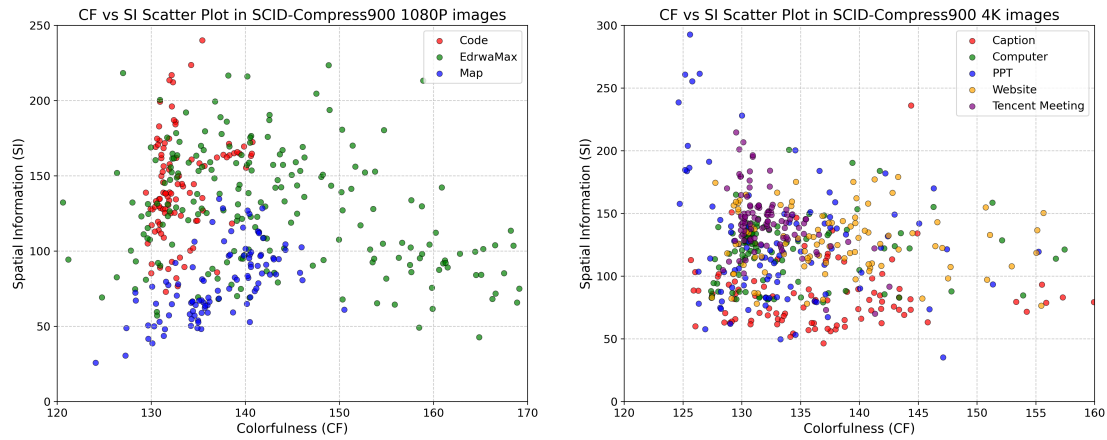


Figure 2: The scatter plots of CF and SI in SCID-Compress900.

Table 1: Comparison of Different Screen Content Image Datasets.

	SCID [9]	SIQAD [18]	CCT [7]	SCID-Compress900
Total number of source images	40	20	24	900
Maximum resolution	1280×720	826×728	1920×1080	4K
Classified?	No	No	No	Yes
Image format	BMP	BMP	PNG	PNG/BMP
Main application	Image quality assessment	Image quality assessment	Image quality assessment	Image compression

- **Map** category (100 images): Covers map drawing and display interfaces.

The 4K resolution includes 500 images, further divided into 5 categories:

- **Caption** category (100 images): Mainly features subtitle-overlay scenarios.
- **Computer** category (100 images): Presents system desktops and multi-window operation interfaces.
- **PPT** category (100 images): Selects business presentation slide interfaces.
- **Tencent Meeting** category (100 images): Records video conference interfaces.
- **Website** category (100 images): Includes various web browsing interfaces.

This classification and resolution configuration not only satisfies research needs in different scenarios but also accommodates algorithm testing under diverse computing power scenarios. Fig. 1 shows some typical images in SCID-Compress900.

3 Analysis and Benchmarking Test

3.1 Comparative Analysis

Table 1 presents a multi-dimensional comparison of the proposed dataset with several classical screen content image datasets (SCID [9], SIQAD [18], and CCT [7]). In terms of the total number of source images, SCID-Compress900 has 900 source images, significantly more than the 40 in SCID, the 20 in SIQAD, and the 24 in CCT. This abundant data volume provides a rich sample basis for algorithm

training and testing. In terms of resolution, SCID-Compress900 supports 4K resolution, which can simulate screen content images in high-definition and even ultra-high-definition display scenarios. In contrast, the maximum resolution of SCID is only 1280×720 , that of SIQAD is 826×728 , and that of CCT is 1920×1080 , all of which cannot meet the needs of algorithm research and development in the ultra-high-definition era. Content classification is a major feature of SCID-Compress900. The dataset performs detailed classification of images, covering various types of scenarios such as programming interfaces, conference charts, and maps. This feature enables researchers to develop and optimize compression algorithms for different types of screen content images to achieve precise performance improvements. However, SCID, SIQAD, and CCT have not undergone content classification, which limits the adaptation of algorithms to real application scenarios. In addition, while SCID, SIQAD, and CCT are mainly used for image quality assessment, SCID-Compress900 focuses on image compression research, which is more in line with the research direction of efficient screen content compression and provides a more targeted test environment for the optimization of compression algorithms.

3.2 Visual Feature Analysis

In the field of screen content image compression, the development of efficient compression algorithms relies on an accurate grasp of the intrinsic features of images. As two key metrics, Colorfulness (CF) [12] and Spatial Information (SI) [3] play a pivotal role in describing the characteristics of screen content images. CF metric is primarily used to measure the vividness, diversity, and overall color expressiveness of an image. From the perspective of human

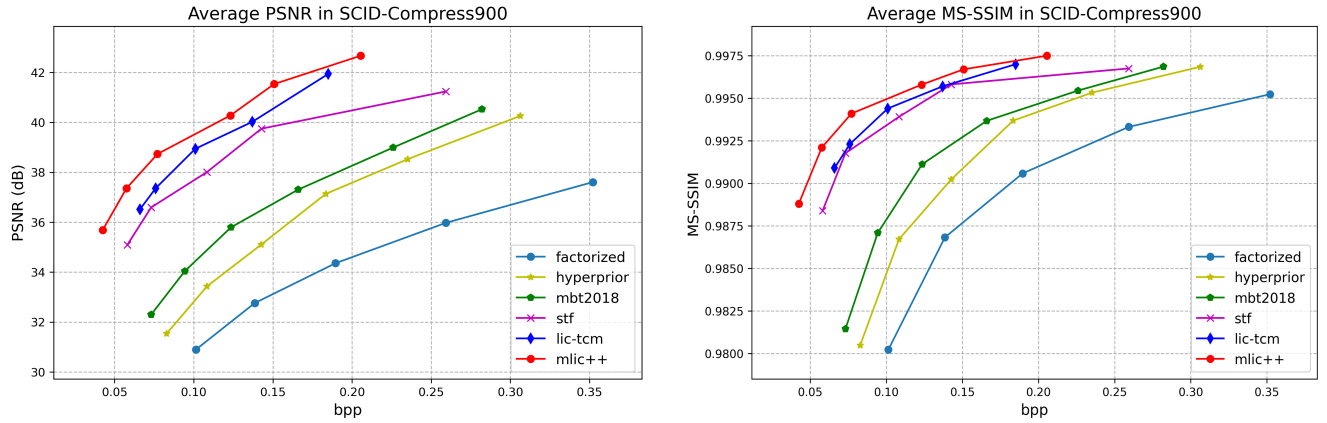


Figure 3: Rate-distortion curves for benchmarking test in SCID-Compress900 dataset.

visual perception, images with rich color are often more capable of attracting attention and leaving a deep impression. SI metric mainly focuses on elements such as the spatial layout of objects in the image, texture details, and edge features. It reflects the complexity and richness of the image in the spatial dimension.

Fig. 2 depicts the CF-SI scatter plot of the SCID-Compress900 dataset. The diversified distribution of scatter points vividly reflects the complex interplay between colorfulness and spatial information across resolution categories. In the 4K subset, scatter points spread widely along CF and SI axes. There is distinct clustering for categories like Caption, Computer, and PPT, showcasing the rich variation in high-resolution content details. The 1080P subset presents a concentrated yet diverse pattern across Code, EdrawMax, and Map. Code is characterized by low SI. EdrawMax covers broad CF/SI ranges. Map displays muted CF/SI values. This unique cross-resolution diversity highlights the dataset's key advantage: it simultaneously captures the high-fidelity complexities of 4K content and the typical structural diversity of 1080P mainstream scenarios. This performance far exceeds single-resolution or limited-category datasets. As a result, SCID-Compress900 serves as an unparalleled benchmark for developing compression algorithms. These algorithms effectively balance color and spatial information preservation across resolutions, ensuring broad applicability to real-world screen content.

3.3 Benchmarking Test

To further verify the practicality of SCID-Compress900 in compression tasks, we conduct benchmarking test on SCID-Compress900 using six deep learning-based image compression algorithms commonly used by researchers (namely factorized [1], hyperprior [2], mbt2018 [8], stf [23], lic-tcm [6], and mlic++ [4]). The evaluation metrics selected are PSNR and MS-SSIM. PSNR is used to measure the pixel-wise error between the original image and the distorted image, while MS-SSIM considers the structural information of the image and human visual perception, making it closer to the subjective quality perceived by the human eyes. The experimental results are shown in Fig. 3. Different algorithms exhibit significant performance differences on SCID-Compress900, indicating that the

dataset we designed has good discrimination in the performance of different algorithms. In addition, the wide coverage of bitrate ranges and large quality variations among different algorithms also reflect the data diversity of SCID-Compress900.

4 Conclusion

In this paper, we establish SCID-Compress900, a new screen content image dataset specifically constructed for screen content image compression research. It consists of 900 high-resolution screen content images, focusing on scenes rich in text and graphics. Through meticulous analysis of the colorfulness and spatial information within the dataset, it is demonstrated that SCID-Compress900 comprehensively captures the characteristics of typical screen content, ensuring its representativeness. We conduct benchmarking tests on commonly used deep learning-based compression algorithms, highlighting its effectiveness in distinguishing algorithm performance. SCID-Compress900 will drive innovation in screen content compression research, facilitate the standardization of compression algorithms in practical applications, and lay a solid foundation for subsequent research on algorithm robustness, generalization capabilities, and other aspects.

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